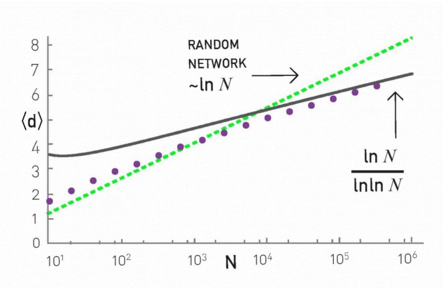
L5: Length of Shortest Paths in PA Model



How about the length of the shortest paths in PA networks? How does their CPL scale with the network size *N*?

It turns out that the CPL in PA networks scales even slower than in *G(n,p)* random graphs. In particular, an approximate expression for the CPL in PA networks is $O(\frac{\ln N}{\ln \ln N})$, which has sub-logarithmic growth. This is shown with simulation results in the plot, where the PA networks were generated using *m=2*.

To summarize, the PA model generates networks with a power-law degree distribution (exponent=3), a clustering coefficient that decreases with network size as $\frac{(\ln N)^2}{N}$, and a CPL that increases sub-logarithmically as $O(\frac{\ln N}{\ln \ln N})$.

Food For Thought

Look at the literature for a derivation of the previous formula for the CPL of PA networks.